

How do different thawing and cooking methods interact to affect the moisture retention (juiciness) of chicken breasts?

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EXPERIMENTAL DESIGN AND ANALYSIS OF VARIANCE

For many college students managing their diets for fitness and health, chicken breast is an indispensable staple. However, preparing it daily often results in dry, unappetizing meals. The question of "How can we cook chicken breast to keep it juicy?" is a highly relatable and trivial culinary dilemma that anyone on a diet has likely faced.

This project aims to address this everyday challenge by applying the rigorous and scientific lens of Design of Experiments. Specifically, we investigate how two common household preparation factors—Thawing Method and Cooking Method—affect the moisture retention (juiciness) of chicken breasts, and whether there is an interaction effect between them.

By doing so, we seek to statistically identify the optimal preparation conditions for the juiciest chicken breast, demonstrating how data-driven scientific methods can effectively resolve even the most trivial problems in our daily lives.

I. Treatments

Our experiment involves a 2×3 factorial design. The treatments are defined as follows:

1) Factor A: Thawing Method (2 levels)

- **Slow Refrigerator Thawing:** Frozen chicken breasts are placed in a household refrigerator for exactly 24 hours until the core temperature reaches $2\sim 4^{\circ}\text{C}$.
- **Rapid Microwave Thawing:** Frozen chicken breasts are defrosted using a 700W microwave. We will continuously check the core temperature until it reaches the target of $2\sim 4^{\circ}\text{C}$.

2) Factor B: Cooking Method (3 levels)

To prevent nuisance variation from different heat transfer rates, the stopping condition for cooking is reaching a core temperature of $75\sim 80^{\circ}\text{C}$.

- **Boiling:** Heat water on an induction stove set constantly to level 8. Once the water begins to boil, submerge the chicken and start the timer. Flip the chicken exactly twice during cooking. Remove when the core reaches $75\sim 80^{\circ}\text{C}$.

- Pan-frying: Preheat a pan on the induction stove at a constant level 8 with 3 tablespoons of olive oil. Place the chicken in the pan and flip it every 30 seconds. Remove when the core reaches 75~80°C.
- Microwaving: Cook in a 700W microwave. Based on our pilot test, achieving the 75~80°C target takes exactly 1 minute. The chicken will be heated for 30 seconds, flipped once, and heated for another 30 seconds.

II. Response Variables

1) Primary Response (Quantitative): Final Weight Loss Percentage (%)

- Academic Rationale: According to [prior poultry science literature](#)¹, cooking loss is defined as the phenomenon where weight decreases due to the loss of moisture and fat during the cooking process. This study highlights that the degree of weight loss varies distinctly depending on the applied heating method, temperature, and cooking time. Therefore, calculating the final weight loss percentage serves as a highly valid, scientifically substantiated primary response variable to evaluate moisture retention (juiciness).
- Instrument: Digital electronic scale
- Procedure: Weigh the standardized raw piece to get the initial weight. After cooking, let the chicken rest at room temperature for exactly 60 seconds to stabilize the juices. Lightly pat the surface once with a paper towel to remove excess oil/water, then measure the final weight.
- Calculation:

$$\text{Weight Loss Percentage (\%)} = \frac{W_{\text{initial}} - W_{\text{final}}}{W_{\text{initial}}} \times 100$$

2) Secondary Response (Qualitative Validation): Sensory Evaluation (Tasting)

To mathematically validate that weight loss accurately acts as a surrogate for "appetizing texture," a sensory evaluation will be conducted. The cooked samples are blinded using random codes. After tasting, the evaluators independently assign scores according to the 7-point Likert scale below, and the arithmetic mean is recorded.

1. 가열기구에 따른 조리방법이 닭고기의 품질특성에 미치는 영향, 한국식품연구원 전기홍 외 5, 한국조리학회지 제 20권 제 3호

Score	Juiciness	Hardness	Textural Preference
7	Extremely Juicy	Extremely Tough	Like Extremely
6	Juicy	Tough	Like
5	Slightly Juicy	Slightly Tough	Like Slightly
4	Neutral	Neutral	Neither Like nor Dislike
3	Slightly Dry	Slightly Tender	Dislike Slightly
2	Dry	Tender	Dislike
1	Extremely Dry	Extremely Tender	Dislike Extremely

III. Experimental Design and Randomization Details

- **Experimental Units:** We use "CJ CheilJedang The Healthy Chicken Breast". To eliminate variance from differing thickness and surface area, the fully frozen chicken breasts are meticulously trimmed into identical geometric cubes measuring 4 cm(W) × 4 cm(L) × 3 cm(H), weighing 30g.
- **Design Structure:** A 2 × 3 Factorial Design with 3 replicates, yielding a total of 18 observations.
- **Randomization & Blocking:** To control for lurking variables, the run order of the 18 trials will be completely randomized. Furthermore, to prevent severe sensory fatigue, the experiment will be divided into 6 days, with the "Day" acting as a blocking factor.

IV. Justification for Factors, Levels, and Responses (Based on Pilot Data)

To validate our experimental design, we conducted a preliminary pilot test focused on benchmarking the preparation times required to achieve identical core temperatures across different methods, ensuring that differences in heat transfer rates do not bias the results.

- **Thawing Benchmarking:** After conducting a 24-hour slow refrigerator thawing process, the core temperature of the chicken breasts consistently measured between 2~4°C. Based on this baseline, we ran pilot trials for the rapid microwave thawing method to determine the exact duration required to reach the identical core temperature of 2~4°C.
- **Cooking Benchmarking:** Our target core temperature for cooking is set between 75~80°C to guarantee food safety and control for overcooking. We conducted pilot runs specifically for the microwave cooking method to find the optimal heating time that satisfies this temperature range, which was determined to be

exactly 1 minute. For boiling and pan-frying, the core temperature can be monitored continuously and measured in real-time during the main experiment; therefore, these two methods proceed directly to the main inquiry without separate pilot benchmarking.

Rationale: The pilot test demonstrates that different appliances require highly variable timeframes to reach identical thermal states. By establishing 2~4°C as the baseline for thawing and 75~80°C as the baseline for cooking, we firmly justify our control mechanisms. This empirical approach ensures that our primary response (Weight Loss %) and secondary response (Sensory Evaluation) are driven solely by the treatment combinations rather than confounding variables like arbitrary cooking times.

V. Power Analysis

To ensure our experiment is statistically adequately powered to detect meaningful differences, we conducted a prospective power analysis using the empirical variance observed in our preliminary pilot testing.

- **Substantiated Inputs and Effect Size:** From our pilot data, the pooled standard deviation (σ) for weight loss across the tested cooking methods was approximately 3.5%. Based on our sensory evaluation scale, we determined that a 5.0%p difference in weight loss translates to a distinctly noticeable 1-to-2 point shift on the 7-point Juiciness scale. Therefore, we established our Minimum Detectable Effect (MDE) at 5.0%p. Using these parameters, the standardized mean difference (Cohen's d) is calculated as $d = 5.0/3.5 \approx 1.42$. Converting this for the ANOVA main effect (comparing two groups, such as the thawing methods), we derive an estimated effect size of Cohen's $f = 1.42/2 = 0.71$.
- **Execution & Conclusion:** Using the pwr package in R, we calculated the statistical power for our 2×3 factorial design with 3 replicates (Total $N = 18$, with $n = 9$ per main effect group). At a significance level of $\alpha=0.05$ and an effect size of $f = 0.71$, the design achieves a statistical power ($1-\beta$) of 0.88. This value comfortably exceeds the conventional academic threshold of 0.80, confirming that our proposed sample size of 18 observations is highly robust and sufficiently powered to detect the main effects without the need for additional replicates.

VI. Ethics and Safety Concerns

- **Food Safety:** To prevent foodborne illnesses associated with undercooked poultry (e.g., Salmonella), a sterilized digital probe thermometer is used to guarantee every sample reaches a safe core temperature of 75~80°C before consumption.
- **Human Participants:** The sensory tasting involves strictly the two group members. Both have confirmed they have no poultry or olive oil allergies. Because there are

no external human subjects or abnormal physical risks, no specific IRB or external ethical measures are required.

VII. Reasonable Analysis Plan (Formal Model)

The collected 18 continuous weight data points and sensory evaluation scores will be analyzed within a Two-way ANOVA framework using R software.

1) Mathematical Formal Model for the Response Variable:

$$Y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \epsilon_{ijk}$$

- Y_{ijk} : The final weight loss percentage(%) of the chicken breast obtained from the k-th replicate under the i-th thawing method and the j-th cooking method.
- μ : The overall mean weight loss percentage across all experimental units.
- τ_i : The main effect of Factor A (Thawing Method) ($i = 1,2$)
- β_j : The main effect of Factor B (Cooking Method) ($j = 1,2,3$)
- $(\tau\beta)_{ij}$: The statistical interaction effect resulting from the combination of both factors.
- ϵ_{ijk} : The experimental error term, which is independently and identically distributed with a mean of 0 and a variance of σ^2 ($\epsilon_{ijk} \sim N(0, \sigma^2)$).

2) Assumption Testing and Hypothesis Verification Procedure:

- Assumption Testing: Prior to the formal analysis, a residual analysis will be conducted to monitor whether the assumptions of normality (Shapiro-Wilk Test) and homogeneity of variance (Levene's Test) are satisfied. If these assumptions are violated, a Box-Cox data transformation will be considered.
- ANOVA Table Derivation: Using the F-test, the statistical significance (p-value) of the interaction effect ($H_0: (\tau\beta)_{ij} = 0$) will be evaluated as the highest priority.

3) ANOVA Table Structure and Degrees of Freedom: The total 17 degrees of freedom will be partitioned into 1 degree for the main effect of the thawing method, 2 degrees for the main effect of the cooking method, 2 degrees for the interaction effect, and 12 degrees for the error term. Allocating 12 degrees of freedom to the error term provides a sufficient basis for a robust estimation of the error variance.

4) Software and Packages: All statistical analyses will be conducted using R software. We will utilize the aov function for model fitting, the car package for Levene's test, and the emmeans package for calculating estimated marginal means and conducting Tukey's post-hoc comparisons.

5) Interpretation of Interaction and Post-hoc Testing: If a significant interaction effect is detected, we will carefully examine the interaction plot to determine if it is an ordinal or a crossover interaction. This will dictate whether the main effects can be interpreted independently or if the optimal cooking method strictly depends on the specific thawing method used. Subsequently, a Tukey post-hoc test will be performed to identify significant differences among specific thawing and cooking combinations that minimize moisture loss.

VIII. Modifications from the Proposal

We have iteratively refined our experimental design by incorporating the feedback received on our initial proposal to make the execution plan more fail-proof:

- **Strict Control of Geometric Thickness:** Our preliminary proposal specified controlling only the weight of the chicken breasts. Following the feedback that thickness heavily influences the thermal response and moisture retention, we introduced a strict geometric preparation protocol. All experimental units are meticulously trimmed into identical 4cm × 4cm × 3cm cubes. This structural standardization eliminates latent nuisance variation and ensures uniform heat penetration across all treatments.
- **Addition of Sensory Validation (Tasting):** To verify that our primary quantitative response (Final Weight Loss Percentage) acts as a reliable surrogate for "appetizing texture"—which is the ultimate culinary objective of this study—we integrated a formal sensory evaluation step. As detailed in Section 2, the experimenters will conduct a blind tasting to assess the juiciness, toughness, and overall textural preference using a 7-point scale.
- **Paradigm Shift to Temperature Benchmarking:** Through our preliminary pilot runs, we discovered that fixing the cooking "time" across different methods causes severe overcooking or undercooking errors due to varying heat transfer rates. Consequently, we changed our control mechanism to target a precise core temperature of 75~80°C as the stopping condition, significantly improving experimental precision.

IX. Timeline for Remaining Work

Current Progress: As of today, the preliminary pilot testing and the first half of the main data collection (9 out of 18 runs) have been successfully completed.

- May 26: Submit the Project Progress Report on Snowboard.
- May 27 – May 29: Complete the remaining half of the main experiment (9 runs). To prevent experimenter palate fatigue during the sensory evaluation, we will strictly conduct 3 runs per day over 3 days, finalizing all data collection by the end of this week.
- June 1 – June 5 (1st Week of June): Data entry, data cleaning, and formal statistical analysis (Two-way ANOVA, assumption testing, and post-hoc analysis) using R software.
- June 8 – June 12 (2nd Week of June): Draft the Final Report and design the Final Presentation slide deck based on the statistical conclusions.
- June 13 – June 14: Final review of all project deliverables to ensure they meet the academic rubrics before the final submission.